

MONALISA DOKANIA

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PROFESSIONAL SUMMARY

Founding Software Engineer with experience in full-stack development, scalable/reliable backend services, distributed systems, data-driven pipelines, and scalable cloud platform design. Curious & motivated learner, with a strong sense of ownership.

TECHNICAL SKILLS

Languages	Python, Java, Go, TypeScript, JavaScript, HTML, CSS
Frameworks / Distributed Syst	React, Node.js, tRPC, Flask, FastAPI, REST API, Spring Boot, Microservices
Cloud & DevOps	AWS, Kubernetes, Jenkins, Docker, Terraform, CI/CD, Kafka
Databases	PostgreSQL, NoSQL, Redis, MySQL, Elasticsearch, MongoDB, data modeling
Software Engineering	SDLC, Design Patterns, System Design, System Architecture, Agile
Machine Learning & AI	Feature engineering, ML model prototyping, AI-assisted workflows, LLM frameworks

TECHNICAL EXPERIENCE

Lead Software Engineer BLUSVN Technologies

June 2024 – Present
Scottsdale, AZ

- **Founding engineer** on a distributed water-delivery platform; **building from 0 to 1**, and owning the full product lifecycle from architecture through production across backend, infrastructure, and ML.
- Architected and shipped **100+ TypeScript/tRPC backend APIs** powering driver and customer React apps, including identity and auth workflows; Improved data-fetching efficiency and reduced end-to-end latency by **40%**.
- Engineered high-performance **Golang microservices** with concurrent worker pools and batched notification delivery, designed for high-throughput driver allocation and order processing at scale.
- Built real-time **ML inference pipelines** for face and ID validation, and **AI-assisted agent workflows** (multi-agent system) for log analysis that cut manual triage time on production incidents.
- Designed and operated production **AWS/EKS infrastructure** with Terraform; migrated from Cluster Autoscaler to Karpenter for faster node provisioning and reduced AWS spend by **12%**.
- Hardened security posture with AWS/EKS best practices - cut high-severity CVEs by **20%**, cluster compliance from **65% to 76%**.
- Built **Kafka-based** event-driven location workflows to support responsive UI experiences in order & delivery status tracking.
- Led **load and chaos testing** with Locust and fault-injection drills, surfacing bottlenecks and validating SLOs before production rollout - validated the platform at **50K orders/min**, and monitored via Prometheus + Grafana observability stack.
- Led engineering delivery as Scrum Master — managed CI/CD pipelines, reviewed 200+ PRs, mentored developers and interns, and partnered with product on technical architecture.

Graduate Teaching Assistant (GTA/GSA) Arizona State University (SCAI)

May 2023 - May 2024
Tempe, AZ

- Supported Software Verification, Validation Testing coursework for large classes (900+ students total).
- Guided students in unit testing, debugging, code reviews, software reliability and object-oriented design concepts.

OTHER PROJECTS

Agent-Based AI System for Document & Log Analysis 🐙 | Python, LangChain, FastAPI, Redis RQ, Vector DB, OpenAI

- Built a custom **multi-agent LLM analysis pipeline** (Planner → Retriever → Analyst → Critic → Reporter) to perform structured, evidence-backed document and log analysis, reducing hallucinations compared to single-prompt workflows.
- Implemented a RAG workflow for **semantic document retrieval** and raw log diagnostics using **FastAPI, Redis RQ, and Qdrant vector store**, enabling asynchronous analysis of large PDFs and logs with verification and confidence scoring.

EDUCATION

Master of Science, Computer Science
Arizona State University | Tempe, AZ

GPA: 4.0/4.0

Bachelor of Technology, Information Technology
Motilal Nehru National Institute of Technology | Allahabad, UP, India

CGPA: 7.59/10.0